

Birmingham City University



DIG5111 Digital Signal Processing

Tutorial 6: Spectral Analysis with the DFT

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Task 1: Spectral Analysis

Look at the *plotting_spectra.m* file in this session's materials. This is a quick overview of how to plot magnitude and phase spectra using MATLAB.

Using the *fft()* function in MATLAB, investigate the magnitude and phase spectra of various signals:

- Sinusoidal signals.
- Sums of sinusoids.
- Musical signals.

Plot the spectra in some different ways:

- Plot the magnitude in decibels: $\text{dB} = 20 \log_{10}(A)$
- Logarithmically scale the frequency axis using the *semilogx()* function.

Task 2: Spectral Processing

Look at the *spectral_processing.m* file in this week's materials. This provides a basic explanation of how to process a signal in the frequency domain. In brief, the process is as follows:

1. Take the DFT using *fft()*.
2. Manipulate the DFT coefficients in some way. Remember that the k^{th} coefficient represents the frequency $\frac{k f_s}{N}$.
3. Take the IDFT using *ifft()*. Using the *symmetric* flag in the call to *ifft()* means you only need to process the positive frequencies.

Alter this code to apply some other spectral effects: e.g. boost some frequencies, alter the phase of certain frequencies.

Task 3: Calculating the Discrete Fourier Transform (DFT)

The definition of the DFT is:

$$X[k] = \sum_{n=0}^{N-1} x[n] e^{-\frac{2\pi jkn}{N}}$$

for $k = 0, 1, 2, \dots, N - 1$

Where x is the input sequence, N is the length of that signal, and X is the sequence of DFT coefficients.

Write a MATLAB function which implements this formula for finding the DFT of a given sequence. Remember that in MATLAB the first element in an array has index 1: you will need to add 1 to the value of n or k when using them to index an array.

Compare the speed and accuracy of your implementation against the MATLAB *fft()* function.

Task 4: Chirp Signals and Spectrograms

The formula for a linear chirp signal is:

$$m = \frac{f_1 - f_0}{2T}$$

$$x(t) = \cos(2\pi(mt^2 + f_0t))$$

Where f_0 is the start frequency, f_1 is the end frequency, and T is the time taken to sweep between the two.

Write a function to synthesise linear chirp signals. With the generated chirp signals:

- Plot the magnitude spectra:
- Use the *stft()* or *spectrogram()* functions in MATLAB to apply time-frequency analysis.

What does each type of analysis tell you about the signal? How does changing the parameters of the STFT (window size/type, overlap) change the analysis?